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PROJECT MANAGEMENT WITH IFC: TOWARD AN OPEN-BIM FRAMEWORK

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Abstract

Successful construction management relies on interoperability, cost estimation, and scheduling. Current practices suffer from fragmented processes and data loss, due to proprietary silos and lack of automation. This research proposes a computational framework to optimize 4D and 5D BIM workflows by enriching the IFC schema. Through *IfcOpenShell*, the methodology provides semantic validation via IDS and collaboration via BCF, facilitating extraction of properties and quantities, scheduling, cost estimation, and resource allocation. Results indicate that an integrated OpenBIM framework using IFC as the primary source improves reliability. The contribution is a reproducible, open-source framework providing a methodology for multi-dimensional BIM management.

Introduction

Building Information Modelling (BIM) transformed the Architecture Engineering Construction (AEC) industry from a static 2D representation to a dynamic parameterized data environment. Despite the methodological maturity suggested by international standards like ISO19650:2018 series, a significant "BIM gap" persists between theoretical frameworks and professional practice (Abanda et al., 2025). The industry relies on closed-proprietary software that creates rigid sequential dependencies that restrict interoperability (Lou et al., 2021) and facilitate the propagation of human errors (Sacks et al., 2018). Integrating BIM with process simulations can address these uncertainties, transforming the model into a decision-support environment capable of mitigating delays and overcoming the limitations of closed systems (Wang et al., 2014). Managing complex 4D and 5D datasets within these environments often requires intensive visual programming (Lou et al., 2021), such as *Dynamo*, to handle the overwhelming volume of repetitive tasks inherent in Quantitative Take Off (QTO) and Bill of Quantities (BoQ). Furthermore, empirical evidence suggests that BIM-based QTO is still rarely applied in practice due to the complexity of the software and incomplete data extraction methods (Alathamneh et al., 2024). The effectiveness of these workflows is inherently linked to the level of information needed which defines the usefulness of the model throughout the life-cycle (Pinti and Bonelli, 2022). In this context, the granularity of information is crucial to calibrate data density and prevent data gaps and over-modeling (Bolpagni and BIMe Initiative, 2016). Interoperability is facilitated by Industry Foundation Class (IFC), which provides machine interpretability of information and allows automation of workflows (Avogaro and Zanchetta, 2025). For project management purposes, entities such as *IfcWorkPlan*, *IfcWorkSchedule* and *IfcTask* regulate temporal aspects, while *IfcCostSchedule*, *IfcCostItem* and *IfcCostValue* provide the foundation for economic estimation. Recent advances, such as Information Delivery Specification (IDS) and BIM Collaboration Format (BCF), have further enhanced the ability to define computer-interpretable requirements and streamline model-based coordination (Fischer et al., 2025). To address these systemic inefficiencies, this research investigates the transition toward OpenBIM computational framework (McGinley et al., 2023). Although current IFC-based exchanges support object-level data, the integration of automated 4D and 5D logic remains insufficiently explored outside of proprietary silos (Avogaro and Zanchetta, 2025). This study proposes a methodology to reliably extract, manipulate, and validate project data by leveraging open-source logic. By bypassing the limitations of proprietary ecosystems, the research proposes a methodology toward the "Total BIM" (Disney et al., 2022) vision, where the digital model serves as the primary source of truth for contractual and management purposes (Pinti and Bonelli, 2022). The paper is organized as follows: Section 2 provides a Literature Review of the IFC schema and construction management. Section 3 and 4 detail the Materials adopted and Key Concepts of the framework, while Section 5 describes the implemented Methodology. Finally, Section 6 and 7 present the Discussion of results and the Conclusion with future research directions.

Literature Review

IFC and Construction Management

The IFC, currently an ISO16739-1:2018 with *IFC4 ADD2 TC1*, provides an extensible, vendor-neutral schema designed to facilitate semantic interoperability within the AEC industry. Unlike proprietary formats, the IFC structure is organized into a four-level hierarchical architecture (Resource, Core, Interoperability, and Domain), allowing for the digital representation of complex construction domains, including actors, processes, and resources.

In the context of construction management, the schema lies in its ability to establish an explicit *IfcRelationship* between the physical component of the building, *IfcProduct*, and the management-related entities. This study exploits the ontological robustness of the IFC schema to synchronize three critical domains: temporal, economic, and resources-based.

IFC, IDS and BCF

The IDS represents a standard that defines Information Requirements (IR) as both human-readable and computer-interpretable (Fischer et al., 2025). The main purpose is to overcome the traditional limitations of exchanging requirements through non-interoperable formats, such as PDF documents, which often hinder automation and data accessibility. By specifying entity types, properties, and classifications, IDS enables automated compliance checks, ensuring that the IFC model contains the semantic richness necessary for 4D and 5D analysis (Fischer et al., 2025). For example, a gap addressed by design firms and public administration is the practical implementation of IR through IDS, particularly in linking Work Breakdown Structure (WBS) to IFC entity (deMarco et al., 2024). While the IFC schema defines the data model, the BCF standard serves as a model-based communication protocol (Lou et al., 2021). By linking IFC entities with BCF topics, stakeholders can communicate and collaborate without sharing the entire model, establishing traceability through *GlobalId*.

IFC and QTO

Despite the potential of BIM-based workflows, a recent systematic review suggests that QTO is still predominantly performed using 2D methods due to software complexity, modeling effort and data inconsistency (Alathamneh et al., 2024). The effectiveness of QTO depends on consistent data and standardization of information exchange (Pinti and Bonelli, 2022). IFC provides a concrete framework for defining measurable elements through *IfcQuantitySet* (Karlshøj and Proctor, 2025). The specification distinguishes between Base Quantities, which provide standardized geometric measures for early-stage estimation, and Custom Quantities, which allow the integration of local rules of measurement or specific professional requirements.

IFC and Project Scheduling

IFC provides a solution to data fragmentation by providing a standardized schema for time information (Xue et al. (2015); Avogaro and Zanchetta (2025)). *IfcWorkPlan* represents the central 4D entity, structuring all the information needed to complete complex activities. The scheduling hierarchy is managed through *IfcRelAggregates* which decomposes *IfcWorkPlan* into specific *IfcWorkSchedules*, and *IfcRelAssignsToControl* which links temporal elements directly to physical *IfcProduct* components. Individual activities are instantiated as *IfcTask* entities, organized through *IfcRelNests* and characterized by temporal data via *IfcTaskTime*. Logical dependencies are established through *IfcRelSequence*, while *IfcRelAssignsToProcess* maps *IfcProduct* instances to their corresponding *IfcTask* anchoring the physical geometry of the digital twin to the temporal workflow. *IfcWorkCalendar* defines standardized working (*WorkingTime*) and non-working (*ExceptionDate*) periods, providing precise control over working-time constraints and supporting accurate sequencing, thus strengthening the connection between temporal and physical dimensions in the 4D BIM environment (Noardo et al., 2022).

IFC and Cost Estimation

IFC schema supports financial information, through entities such as *IfcCostSchedule* for structuring, *IfcCostItem* for element-based assignment, and *IfcCostValue* for price definitions (Pishdad and Onungwa, 2024). Current research utilizes these capabilities to overcome the limitations of parameter-based 5D workflows. Cassandro et al. (2024a) proposed a methodology implemented through *IfcOpenShell* library that used *IfcCostItem* to establish a direct relationship between costs and IFC geometric entities, proving to be more effective than the 5D BIM approach based on simple assignment of parameters and attributes. The approach was validated in a second study on a structural model, demonstrating its effectiveness in optimizing cost estimates for public tenders and minimizing human error by anchoring *IfcCostItem* instances to official price lists (Cassandro et al., 2024b). *IfcCostSchedule* organizes *IfcCostItem* entities into a coherent hierarchy through *IfcRelAggregates*, ensuring the validity of financial data throughout the building's life-cycle. The entity supports hierarchical nesting through *IsNestedBy* and *Nests* relationships, ensuring that total costs remain the verifiable sum of subordinate items Cassandro et al. (2024a). *IfcCostItem* is linked to one or more *IfcCostValue* instances to define price breakdowns across materials, equipment, and labor, providing a native foundation for BoQ generation that current practice consistently replaces with external spreadsheet-based workflows.

IFC and Resource Allocation

Resource allocation is critical for successful project management, as it directly influences site safety, spatial coordination, and the feasibility of both schedules and budgets. Within the IFC schema, *IfcConstructionResource* provides the formal structure defining activity resources (Avogaro and Zanchetta, 2025). The schema supports complex resource containers through *IfcCrewResource* and *IfcSubContractResource*, which aggregate elementary resources - *IfcLaborResource*, *IfcConstructionEquipmentResource* and *IfcConstructionMaterialResource* - into functional units through *IfcRelNests*. By linking resources to specific tasks and costs, the schema provides the documentation necessary to justify expenses, verify work progress, and maintain a historical record of resource performance, which is important for transparent project reporting and

future cost benchmarking (Pinti and Bonelli, 2022).

Research gap

Despite the ability of the IFC schema to support construction data management, current studies often treat model validation, project scheduling, and cost estimation as isolated processes (Avogaro and Zanchetta, 2025). Despite the expressive power of its scheduling, cost, and resource entities, these native capabilities remain largely underutilised in practice, consistently replaced by proprietary tools that fragment data across disconnected pipelines and never write information back into the schema. Recent contributions have successfully standardized IDS and cost estimation processes, demonstrating that machine-readable data is essential to reduce human errors and legal disputes in public tenders (Cassandro et al. 2024a; 2024c). However, a significant gap remains regarding the dynamic synchronization of time scheduling, cost estimation and resource allocation within a unified OpenBIM workflow. This fragmentation prevents the AEC industry from reaching the "Total BIM" vision (Disney et al., 2022), where the model serves as the only source of contractual and operational information (Pinti et al., 2022). The lack of a transparent methodology capable of orchestrating these different IFC domains remains the main barrier to a fully automated OpenBIM environment. This study addresses this deficit by proposing an integrated framework that leverages open-source computational logic to synchronize IR with multidimensional project control.

Materials

The proposed framework is developed within Free and Open Source Software (FOSS) ecosystem, with the aim to provide a transparent and accessible alternative to proprietary BIM tools. This choice is aligned with the current academic necessity of "Supporting the OpenBIM transition" (McGinley et al., 2023), ensuring open data and long-term interoperability. Adopting a "decoupled" approach, the framework performs data multidimensional (4D and 5D) management and enrichment, independently of the original authoring software avoiding the "black-box" logic of commercial tools. This ensures full compliance with OpenBIM standards and the ISO16739-1:2018 specification, through controllable and verifiable data manipulation. The core logic is implemented in *Python* (v3.10+) taking advantage of its extensive ecosystem in data science and AEC automation. To ensure maintainability reproducibility, the environment is managed via a *Virtual Environment* (.venv) architecture and a *requirements.txt* protocol, while version control is maintained via *Git*. The framework uses *IfcOpenShell* (v0.8.4) documentation as the core engine providing bidirectional access to IFC data structures and allowing users to navigate model hierarchies and resolve inverse relationships between physical components and their semantic properties (Noardo et al., 2022). The technological stack and the standardized protocols integrated into the framework are summarized in Table 1.

Table 1: Technological stack and data exchange protocols

Category	Component	Function
Rendering	*ThatOpen*, *Three.js*	Browser-based fragment rendering via *WASM* and GPU acceleration.
Validation	*IfcTester*, IDS	Automated compliance check of requirements via *XML* IDS files.
Interoperability	BCF, *JSON*	Issue tracking via *IfcOpenShell* and state persistence.
Analysis	*Pandas*, *NumPy*	IFC transformation into *DataFrames*.
Visualization	*Plotly*	Interactive 4D/5D charting.

Key Concepts

The proposed framework establishes a digital workflow designed to automate the validation and data enrichment of IFC models (Noardo et al., 2022). The process begins with model viewing and the validation, then the framework executes a set of algorithms to integrate 4D and 5D information into the system. The methodology is built upon seven fundamental concepts:

i. IDS and BCF: The framework treats (IR) as machine-readable document, verifying that the model contains the correct classifications and geometric properties before any calculation, through IDS (Fischer et al., 2025). If discrepancies are found, the BCF module enables a traceable feedback loop (Lou et al., 2021).

ii. QTO and Cost Estimation: To ensure accurate budget control, the framework performs an automated Quantity Take-Off. The total project cost is calculated by summing direct costs with indirect components (Liu et al., 2014):

$$C_{tot} = \sum_{i} \sum_{j=1}^{n} (Q_j \times R_j) + C_{indirect} \qquad (1)$$

where Q_j is the quantity extracted from the validated IFC geometry and R_j is the unit price mapped from standardized price lists. Each unit price comprises three cost components: $R_j = R_{mat} + R_{equ} + R_{lab}$ (materials, equipment, labour).

iii. Time Estimation: The framework calculates task durations using two methodologies:

- Synthetic Method: Derived from labour cost incidence (I_{man}) provided by regional price lists ($C_{man} = C_{tot} \times I_{man}$).
- Analytical Method: Based on specific production rates and unit labour costs (UP_{man}) directly associated with quantities ($C_{man} = Q \times UP_{man}$).

The total man-hours (H) are converted into working days (T_g) considering the size of the operative crew (N_{so}): $T_g = H/(8 \times N_{so})$.

iv. Program Evaluation and Review Technique (PERT) Scheduling: To account for construction uncertainty, durations are treated as random variables. The framework calculates the expected duration (T_A) using a beta probability distribution:

$$T_A = \frac{a + 4m + b}{6} \tag{2}$$

where m is the most likely duration (T_g), a is the optimistic estimate ($0.75m$), and b is the pessimistic estimate ($1.75m$). This approach ensures a robust Critical Path Method (CPM) evaluation and the identification of critical path activities.

v. Resource Allocation: Each task is linked to a multidimensional array of resources, including costs, materials, equipment and specialized labour. This exhaustive mapping is critical for site safety management, as it enables the identification of spatial-temporal overlaps between hazardous activities and ensures that safety protocols and required certifications are explicitly associated with each operational unit.

vi. Task Sequencing: Construction activities are sequenced through standard logical dependencies (F-S, S-S, F-F, S-F) and visualized through Gantt chart and *DataFrames*.

vii. Cash Flow: The framework aggregates the results of the 4D and 5D synchronization to perform automated financial forecasting. By setting incomes and expenses, the system generates cumulative cost distributions, allowing project managers to monitor the financial health of the project, identifying potential liquidity gaps and comparing budgets against real-time projections.

Methodologies

The workflow in Figure 1 guides the reader towards the logic of the framework, which illustrates the integration of IFC data validation and collaboration, followed by automated quantity extraction necessary for a dynamic project management phase.

IFC Export and Import

The framework approach begins with input standardization through two modules: export configuration and model import. To ensure data quality, the system generates *JSON* files that translate the *IFC4X3_ADD2* specifications into parameters ready for authoring software. Subsequently, the IFC import module loads the file and initializes the working environment. Using *IfcOpenShell*, the framework performs a structural audit that detects the schema version and verifies the integrity of the file. This step is crucial to confirm that the model is ready for semantic enrichment and 4D and 5D analysis.

3D Visualization Architecture

The 3D visualization module handles high-density IFC data by moving processing to a decoupled client-side pipeline. Data are passed as *NumPy uint8* memory blocks from the *Python* backend to a custom *React* component, bypassing the overhead of text-based encodings. The core engine, powered by *WebAssembly*, *Three.js*, and the *That Open Company* ecosystem, converts the model into *Fragments* to group identical geometries and to optimize *GPU* performance. This approach enables native browser-based rendering, eliminating the need for external BIM viewers. To ensure data persistence, camera matrices are serialized into *JSON*, while session stability is maintained through a memory management logic that purges geometry buffers from *VRAM* during module transitions.

IDS and BCF

The framework ensures data consistency through an automated workflow that bridges the gap between IR and collaborative management. This process relies on IDS to validate the model requirements and BCF to communicate discrepancies. The validation engine utilizes *IfcTester* to ensure that IR are machine-readable and to verify the compliance of the model. By parsing *XML*-based IDS files, the tool validates whether the model contains all the required information without manual intervention. Users can dynamically define rules based on IFC classes and *PropertySet*, which the system validates using a robust lookup strategy via *IfcOpenShell*. Using *ifcopenshell.util.element.get_psets*, the engine extracts granular properties and element attributes, transforming complex model hierarchies into a structured *Pandas DataFrame*. This approach identifies missing properties or invalid values, calculates real-time compliance metrics, and generates visual reports using *Plotly*. To close the coordination loop, the framework integrates a synchronization logic between the 3D viewer and the reporting module. When a non-compliant element is identified, the system captures the current state of the model viewer—including camera projection matrices—as standardized snapshots. These visual cues are associated with the specific validation failure. Leveraging the *IfcOpenShell* BCF module, the framework serializes these discrepancies into a *.bcfzip* archive. This package contains the necessary *markup.bcf* and *project.bcfp XML* structures, allowing the identified issues to be re-imported into any BIM authoring software. This ensures a feedback-based workflow where requirements are not only verified, but effectively communicated for resolution.

Properties and Quantities Extraction

The framework transforms IFC models into structured datasets through a schema-independent extraction pipeline based on the *ifcopenshell.util* module. The module dynamically extracts each *IfcPropertySet* instances attached to *IfcProduct* elements, resolving *IfcRelDefinesByProperties* relationships to capture property names, values, and

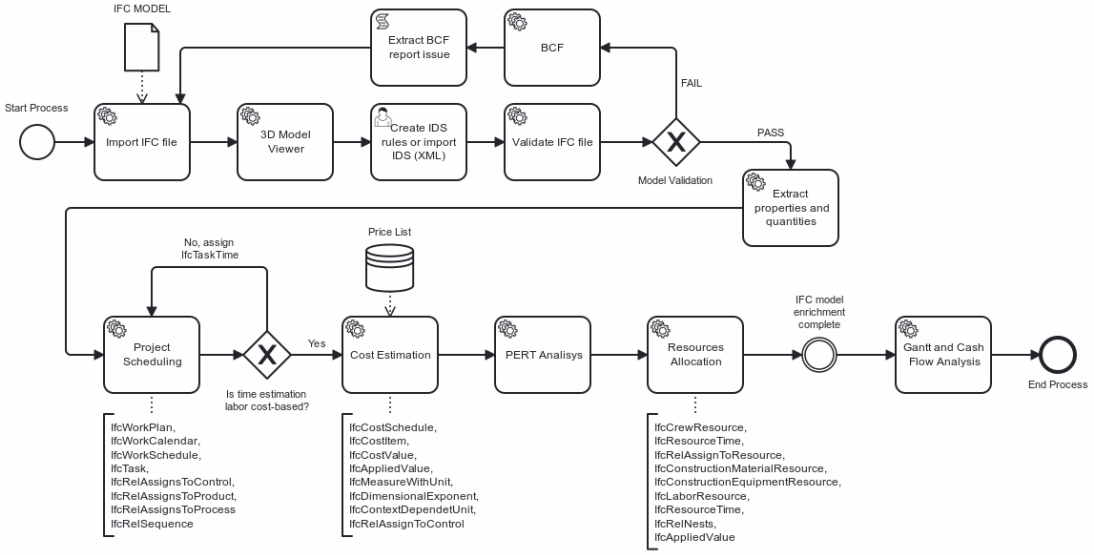


Figure 1: Framework logical process.

data types with full *GlobalId* traceability. This approach ensures compatibility with both standard property definitions (*Pset_**) and custom sets. Geometric quantities are retrieved by parsing *IfcElementQuantity* entities through the inverse *IsDefinedBy* attribute. The extraction logic processes physical quantity subtypes including *IfcQuantityArea*, *IfcQuantityVolume*, *IfcQuantityLength*, *IfcQuantityCount*, and *IfcQuantityWeight*, supporting both standard *BaseQuantities* and domain-specific (*Qto_**) sets. *Pandas* and *NumPy* transform IFC hierarchies into tabular *DataFrames*, enabling efficient grouping, filtering, and quantity calculations for cost estimation and scheduling. Each row represents a property or quantity instance with element identification, classification metadata, source discriminators, and value containers. Furthermore, the extracted quantity incorporates a diagnostic layer that flags potential duplicate entries across element classes. Records that share identical quantity values within the same IFC class are highlighted as potential duplicates. In the test model, for example, querying *NetArea* on *IfcSlab* elements on *F_00 Level* revealed a raw aggregate of $81,807.68\text{m}^2$ against a potential corrected value of $2,586.36\text{m}^2$, demonstrating the capacity to highlight suspicious duplicate quantities of the same element.

4D Scheduling

The temporal dimension is integrated into the model by implementing the IFC schema via the *ifcopenshell.api.sequence* module. The framework establishes a hierarchical structure in which an *IfcWorkPlan*, declared within the *IfcProject* via *IfcRelDeclares*, serves as the main container for multiple *IfcWorkSchedule* entities. Operational constraints are defined via *IfcWorkCalendar*. The framework utilizes *sequence.add_work_time* and *sequence.assign_recurrence_pattern* to specify working hours and recurring exceptions. Individual tasks are instantiated as *IfcTask* entities using *sequence.add_task*. The system supports recursive nesting via *IfcRelNests* via *nest.assign_object*, allowing complex work packages to be broken down into granular subtasks. Each task is associated with an *IfcTaskTime* object via *sequence.add_task*. Through *sequence.assign_process*, the framework identifies which elements are operated upon, while *sequence.assign_product* defines the physical output of each task, defining *IfcRelAssignsToProcess* and *IfcRelAssignsToProduct*. Task sequencing is managed through *IfcRelSequence* relationships created via *sequence.assign_sequence*. The framework supports standard dependency types, enabling dynamic CPM analysis directly within the OpenBIM environment.

5D Cost Estimation and BoQ

The cost estimation framework automates the generation of BoQ by synchronizing costs with the IFC geometric data using the *ifcopenshell.api.cost* and *ifcopenshell.api.control* modules. By importing an external price list (.csv) and instantiating an *IfcCostSchedule*, the framework creates an *IfcCostItem* for each price line using *cost.add_cost_item*, aggregating three distinct *IfcCostValue* entities representing materials, equipment, and labour costs via *cost.add_cost_value*. Each value is associated with an *IfcMeasureWithUnit* and to overcome naming discrepancies between model elements and price list descriptions, the system implements a fuzzy semantic mapping using *difflib.SequenceMatcher*. The pipeline first filters candidate price rows by IFC class using *IfcMatch* field, a classification column manually aligned to the price list structure, retaining only entries compatible with the element type. Within this, *difflib.SequenceMatcher* computes string similarity between the normalized element name and each filtered candidate, and the highest-scoring entry is selected. For example, element *"HEB Søjle_N:HE180B:371924"* - following the convention *FamilyType:TypeName:GlobalId* - cor-

rectly matches *"Stålsøjle HE-B 180"* in the *Molio* price list despite the structural difference (similarity score = 0.47). Once validated, *control.assign_control* establishes the definitive link between the *IfcElement* and its *IfcCostItem*, which are logged and surfaced to the user, who can manually correct erroneous assignments by removing individual *IfcRelAssignsToControl* relationships and retriggering the pipeline. Quantity extraction follows a hierarchical priority logic to ensure accuracy, where net measurements are preferred over gross ones. The framework automatically handles unit conversions by querying *IfcUnitAssignment*. The final economic output is structured as a relational *DataFrame* where costs are filtered and aggregated by *IfcBuildingStorey* using *IfcRelContainedInSpatialStructure* relationships. For each element, the unit price R_j is derived from the sum of the costs of materials, equipment, and labour retrieved from the corresponding *IfcCostItem*, ensuring hierarchical budget breakdowns by level and for the entire project.

Resource Allocation

The resource allocation module integrates construction resources with scheduling by instantiating *IfcConstructionResource* entities and linking them to scheduled tasks via the *ifcopenshell.api.resource* module, supporting *IfcCrewResource* and *IfcSubContractResource*. For each *IfcTask*, the framework determines its *ScheduleDuration* through a PERT analysis in which the total labour hours, extracted from the BoQ, are adjusted based on a defined *crew_size*. To ensure accuracy, the module queries *IfcWorkCalendar* to identify *WorkingTimes* and *ExceptionTimes*. The result is converted into an ISO 8601 duration string and assigned to an *IfcTaskTime* entity through a custom function. Using *resource.add_resource*, a parent resource container is instantiated with attributes, and an *IfcResourceTime* entity is created via *resource.add_resource_time* to store the total labour hours. The parent resource is then assigned to the task via *resource.assign_resource*, establishing an *IfcRelAssignsToResource* relationship. For crew-managed tasks, the parent resource is decomposed into child resources (*IfcLaborResource*, *IfcConstructionEquipmentResource*, and *IfcConstructionMaterialResource*) nested via *IfcRelNests*. Time and financial attributes are embedded in this hierarchy: *IfcLaborResource* instances receive an *IfcResourceTime* object where the *ScheduleWork* attribute reflects the adjusted man-hours, while cost data for all resources is stored via *IfcAppliedValue* entities within the *BaseCosts* attribute. This structure enables a recursive cascade where root tasks and *IfcWorkSchedule* are automatically updated by aggregating the *ScheduleStart* and *ScheduleFinish* dates of their nested sub-tasks, ensuring global time and economic consistency across the IFC model.

GANTT and Cash Flow analysis

The proposed framework processes the previous IFC entities to generate integrated Gantt charts and cash flow visualizations, by extracting task schedules and linking them to cost and resource data. The system reconstructs the project hierarchy by crossing *IfcRelNests* and *IfcRelAggregates* relationships, extracting data from each *IfcTask*. The costs are then mapped to the timeline by intersecting *IfcRelAssignsToProcess* (linking tasks to products) and *IfcRelAssignsToControl* (linking products to cost items), aggregating categorized materials, equipment, and labour expenses. Financial forecasts distribute costs proportionally across task durations rather than treating them as point-in-time events. For example, a multi-week construction activity spreads its total cost across the execution period based on the proportion of work hours in each time window. Processing these data streams through *Pandas*, construction costs are integrated with external incomes and expenses to generate cumulative cost distributions and net cash flow metrics, enabling accurate liquidity planning and early identification of funding gaps, providing a decision support for financial sustainability. This integrated approach establishes a bidirectional link between the IFC model and the project's temporal financial health, fulfilling the "Total BIM" (Disney et al., 2022) requirement of using the digital model as the primary contractual and management record.

Results and Discussion

The results shown in Table 2 demonstrate that the proposed framework bridges the gap between static IFC models and dynamic project management. By automating the "validation loop", only high-quality data populate the model, reducing the manual effort required for compliance checking. Unlike proprietary workflows, the proposed framework operates entirely within the native IFC schema; all scheduling, cost, and resource data are co-located within a single model and semantically linked through native IFC entities. This ensures that any design or on-site change propagates consistently, reducing the operational overhead of maintaining data consistency across multiple proprietary tools. Enriching 4D and 5D entities directly ensures that all project management data remain queryable, portable, and auditable, without requiring any software outside the open-source eco-system overcoming the fragmentation that OpenBIM tries to prevent (Sacks et al., 2018). The proposed framework provides immediate visibility into cost and schedule changes, and by preventing the "cascade effect" of unexpected expenses, the system enables reliable decision-making that protects the quality and overall performance of the project. However, significant challenges and limitations have been revealed in the implementation phase. In particular, the integration of 3D visualization components highlights the gap between BIM data and graphical engines. Furthermore, to overcome current gaps in the *IfcOpenShell* library with respect to specific modules, the framework implements custom algorithms to

Table 2: IFC 4D and 5D main enrichment contribution

Category	Enriched IFC Instance
Scheduling	#78807=IFCWORKPLAN('0NsExtC2f8D8WeUQZlTNzR', #78805, 'Construction WorkPlan', $, '', 'WP-01', '2026-01-14T15:40:00', (#15), $, $, $, '2026-01-14T08:00:00', '2027-01-14T17:00:00', .PLANNED.);
	#78811=IFCWORKCALENDAR('07lFbCthz3nwQka1zjY40E', #78805, 'Construction WorkCalendar', 'WC-01', $, $, (#78812), (#78815), .FIRSTSHIFT.);
	#78813=IFCRECURRENCEPATTERN(.WEEKLY., $, (1,2,3,4,5), $, $, $, $, (#78814));
	#78819=IFCWORKSCHEDULE('0menkKZ$51oAc_AcvL_qld', #78805, 'Construction WorkSchedule', $, $, 'WS-01', '2026-01-14T15:40:00', (#15), $, $, $, '2026-01-14T08:00:00', '2027-01-14T17:00:00', .PLANNED.);
	#78821=IFCRELNESTS('2zVMrPaEL37ex7xLms7nHC', #78820, $, $, #78811, (#78807, #78819));
	#78829=IFCTASK('1NN1H7z_LFCwmdHctg15XD', #78828, 'B_00 Slab', 'B_00 Slab Construction', $, '1.1', $, $, $, .F., 0, #79750, .NOTDEFINED.);
	#78835=IFCRELASSIGNSTOPROCESS('2JPu4ZsZn3OhG98crHGOnU', #78834, $, $, (#46908), $, #78829, $);
	#79005=IFCRELSEQUENCE('1BqjPS6fn74PiTOQRd4ylY', #79004, $, $, #78829, #78833, $, .FINISH_START., $);
Cost	#79007=IFCCOSTSCHEDULE('25qTTCIEzEcgVMOqXhCm3F', #78805, 'Price List Cost Schedule', $, $, $, .SCHEDULEOFRATES., $, $, '2026-01-14T15:48:31.239086');
	#79009=IFCCOSTITEM('10bAGvkHHEw8R1PlUeSNlN', #79008, 'Betonbj\X2\00E6\X0\lke 200x300mm', $, $, '04.10.82,01', $, (#79012, #79016, #79020), $);
	#79011=IFCRELASSIGNSTOCONTROL('0jRM67LPX9Ue3QN_$YIzt6', #79010, $, $, (#79009, …, #79707), $, #79007);
	#79012=IFCCOSTVALUE('Materials', $, IFCMONETARYMEASURE(4056.05), #79015, $, $, 'MATERIALS', $, .MULTIPLY., $);
	#79016=IFCCOSTVALUE('Equipment', $, IFCMONETARYMEASURE(2520.36), #79019, $, $, 'EQUIPMENT', $, .MULTIPLY., $);
	#79020=IFCCOSTVALUE('Labor', 'Labor Hourly Rate: 484.57 c/h', IFCMONETARYMEASURE(15113.7383), #79023, $, $, 'LABOR', $, .MULTIPLY., $);
	#79723=IFCRELASSIGNSTOCONTROL('0j$nKjAVz19OQC8faTeGYg', #79722, $, $, (#17791, …, #63008), $, #79009);
Resources	#79753=IFCCREWRESOURCE('1a_wSbOMz7EhoFL4luGDWg', #79752, 'Crew Resource for Task B_00 Slab', 'Management Type: Own crew...', $, 'RES-B_00 Slab-CRW', $, #79754, $, $, .NOTDEFINED.);
	#79754=IFCRESOURCETIME($, $, $, 'PT15H6M37S', $, $, $, $, $, $, $, $, $, $, $, $, $);
	#79756=IFCRELASSIGNSTORESOURCE('2iXBk4EBP3HAHE$_XGUVrV', #79755, $, $, (#78829), $, #79753);
	#79758=IFCCONSTRUCTIONMATERIALRESOURCE('26lXNDGc55C9z8aufxDE$L', #79757, 'Material - 78829-IfcSlab', 'Type: CONCRETE', $, 'MAT-78829-IfcSlab', $, $, (#79761), $, .CONCRETE.);
	#79760=IFCRELNESTS('3PGGyYVVXB4uYwydQvpI6Q', #79759, $, $, #79753, (#79758, #79763, #79767));
	#79763=IFCLABORRESOURCE('01D3bhwnj7POQu2zvO5Sdl', #79762, 'Labor - 78829-IfcSlab', 'Crew Size: 2 workers', $, 'LAB-78829-IfcSlab', $, #79765, (#79764), $, .CONCRETE.);
	#79767=IFCCONSTRUCTIONEQUIPMENTRESOURCE('2feoeXk_vAPBHTq8qrIptH', #79766, 'Equipment - 78829-IfcSlab', 'Type: PUMPING', $, 'EQP-78829-IfcSlab', $, $, (#79768), $, .PUMPING.);

manage specific hierarchical scheduling complexities. In addition, the intricate relationships and nesting of resource entities made the attribute manipulation complex. While *difflib.SequenceMatcher* provides a scalable baseline for semantic mapping, the test model revealed a structural limitation: across 1 006 elements assigned scores ranged from 0.12 to 0.51 (avg. 0.32), confirming that the primary safeguard is the IFC class pre-filter, while within-class ambiguity requires manual verification. This suggests that integrating standardized classification systems, such as WBS codes or Uniclass classification, would mitigate string similarity inaccuracies and improve reliability. Finally, while *GlobalId* instability across re-exports remains a known OpenBIM challenge, the framework ensures data persistence by establishing associations within a single pipeline, provided that the downstream workflow preserves native IFC entities.

Conclusions

This research proposed an integrated OpenBIM framework designed to transform digital models into active decision-support tools for integrated construction management. Responding to the operational workflows of design firms, public administrations, and construction site managers, the framework bridges regulatory requirements and construction practice. Its main contribution lies in its ability to mitigate repetitive tasks associated with manual data entry, often the main cause of human error and project delays. By providing an accessible alternative to expensive proprietary software, the framework makes high-level BIM management feasible even for smaller organizations and public administrations. Despite these advances, the framework has limitations: it assumes a single authoritative IFC file as its primary data source and complex architectural and Mechanical, Electrical, Plumbing (MEP) models could face challenges with highly complex geometries or intricate systems. The granularity of the data and non-linear relationship between components in these systems challenge data extraction and quantification, requiring more specialized analysis rules. Based on these findings, work is underway to evolve this framework into a user-friendly standalone tool. While qualitative validation is currently being conducted to assess its impact on professional workflows and its effectiveness in diverse project environments, quantitative validation will be conducted to measure the framework's performance in terms of time-saving and error reduction compared to traditional manual methods. Future implementations will address the above limitations and may extend to versioning control, IoT-based site monitoring, and Life Cycle Assessment (LCA) integration. The use of OpenBIM standards ensures high scalability and reproducibility, while the use of *Python*-based technologies ensures the framework's implementation in cloud environments, facilitating collaborative management. Furthermore, to enhance user accessibility, the implementation of a Natural Language Processing (NLP) interface is envisioned. This would allow non-expert users to query the IFC model in natural language (Nabavi et al., 2023).

Data availability

Data are provided under a *Demonstration-Only Proprietary License* and will be made available on request.

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